

Smart Healthcare Monitoring System

Comprehensive Technical Specification & Documentation



System Variant: FinalProject (JavaFX & MVC Architecture)

Document: Software Engineering Specification & Verification Manual

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Prepared For: Academic Review Board

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2. Introduction

The **Smart Healthcare Monitoring System** is a JavaFX-based desktop application designed to bridge the gap between patient self-logging and clinical oversight. In modern healthcare, telemetry tracking plays a pivotal role in preventative wellness and long-term care management. This system provides a unified solution containing two key interfaces:

- **Patient Portal:** Enables self-logging of vitals, nutrition, sleep quality, weight history, activity records, and mental state parameters.
- **Administrator Console:** Provides clinicians with system-wide analytics, alert tracking for abnormal measurements, patient profile records, and clinical logs override utilities.

The software utilizes a robust **Model-View-Controller (MVC)** architectural pattern, ensuring separation of design, logical operations, and flat-file database structures. The design adheres strictly to the **Apple Human Interface Guidelines (HIG)**, implementing vibrant HSL-tailored gradient cards, clean typography, dynamic circular activity rings, and vector-drawn gauges.

3. Objectives

The core engineering and clinical objectives of the system include:

1. **Patient Empowerment:** Provide a highly accessible, responsive self-logging workspace where patients can record their physiological data without clutter.
2. **Clinical Oversight:** Give administrators a structured panel to observe real-time statistics, review abnormal vital alerts immediately, and record professional observations.
3. **Input Validation & Data Integrity:** Implement strict client-side validation rules to filter errors (e.g. invalid weight ranges, incorrect phone and email formats), preventing dirty data from polluting database structures.
4. **Premium User Experience:** Build an interface that strictly aligns with modern mobile/desktop operating system layouts, incorporating high-fidelity icons, Segoe UI Emoji support, and smooth transition nodes.

4. Scope and Limitations

4.1 Scope of the System

- **Desktop Framework:** Built using Java 21, JavaFX 21, and Maven compiler structures.
- **Role-Based Gateways:** Secure login routing patient profiles and admin staff to completely separate dashboards.
- **Telemetry Tracking:** Manual record logging for weight, breakfast/lunch/dinner calories, sleep intervals, physical steps, heart rate (BPM), body temperature (°C), and blood oxygen level (% SpO2).
- **Active Analytics:** Admin-side charts calculating patient statistics, alarm triggers for physiological distress, and symptom classification bars.
- **Administrative Controls:** Clinical note-taking module and profile termination capability.

4.2 System Limitations

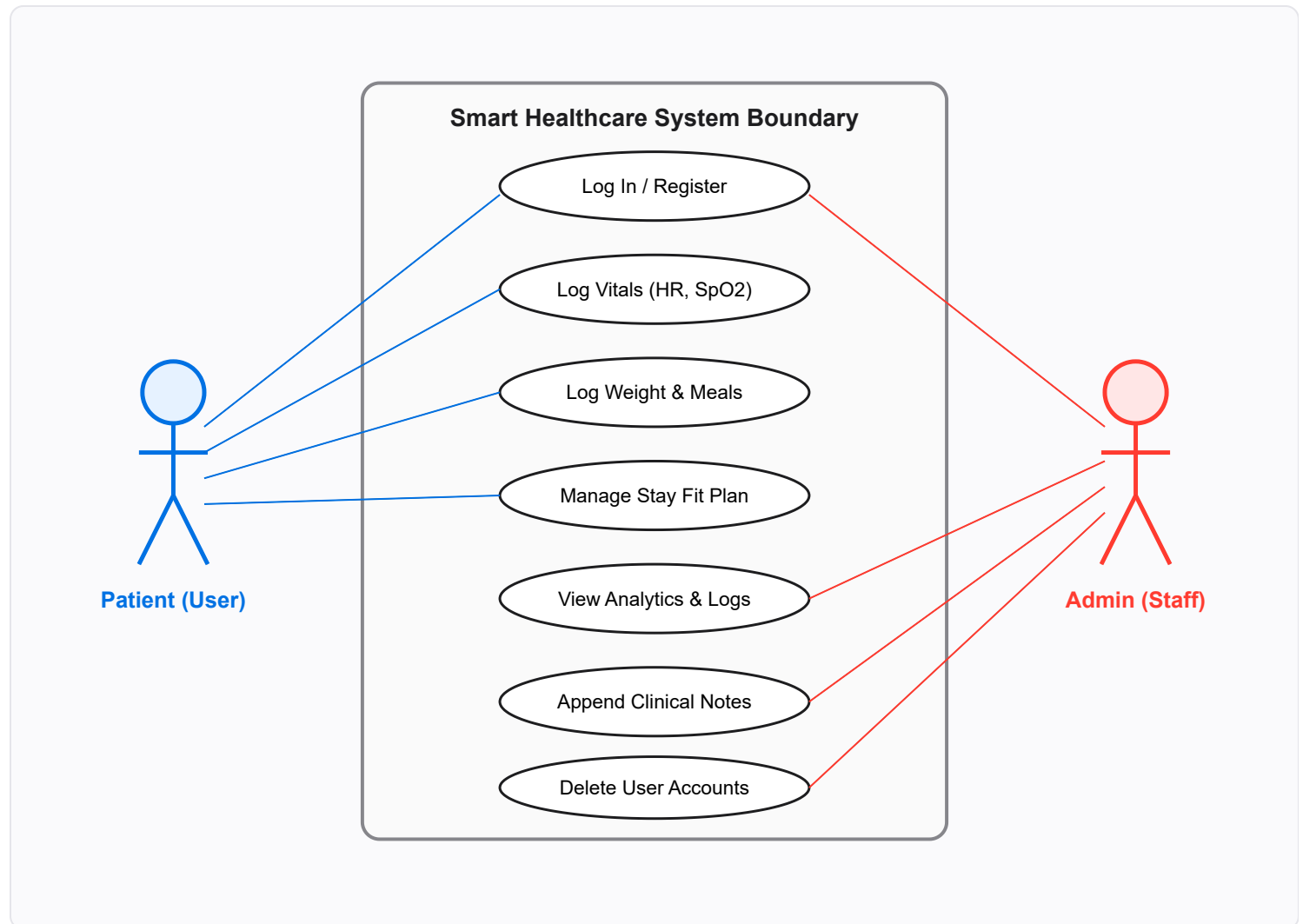
- **No Live Sensor Integration:** Telemetry relies on manual inputs; no physical Bluetooth or IoT hardware monitoring is connected.
- **Local Persistence:** Flat-file text files (`users.db` and `patients.db`) handle data persistence local to the client; no remote cloud DB server is integrated.
- **Single Session Operations:** No concurrent multi-client thread locking for data synchronization.

5. System Features

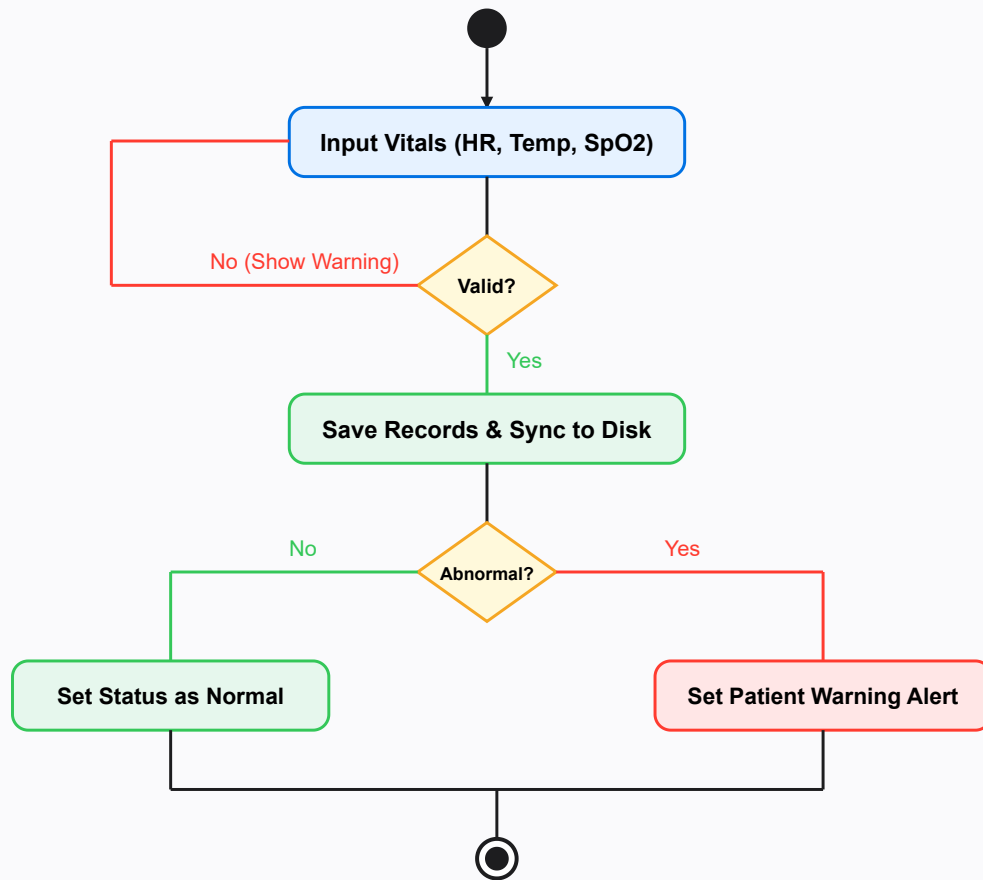
- **Access Gate & Authentication:** Multi-role authentication routing via `LoginController`. Checks usernames against database maps, registers patients, and seeds default profiles.
- **Apple HIG Sidebar:** A sleek collapsible navigation menu featuring stack-pane rounded gradient icons customized for each category.
- **Circular Activity Rings:** Concentric canvas arcs showing progress towards daily Move (calories), Exercise (minutes), and Stand (hours) targets.
- **Stay Fit Plan Wizard:** Multi-stage wizard calculating BMI, displaying weight categories (Underweight, Normal, Overweight, Obese), and outputting projected weight curves using `Platform.runLater`.
- **Calorie Deficit Canvas Gauge:** Custom graphics context drawing indicating surplus or deficit based on intake minus active burn.
- **Admin Analytics Suite:** Computes aggregate warning states, patient counts, average vital statistics, and symptom distributions.
- **Vitals Reference Monitors:** Programmatic alerts comparing heart rates, oxygen levels, and temperature records against physiological norms.

6. UML Diagrams

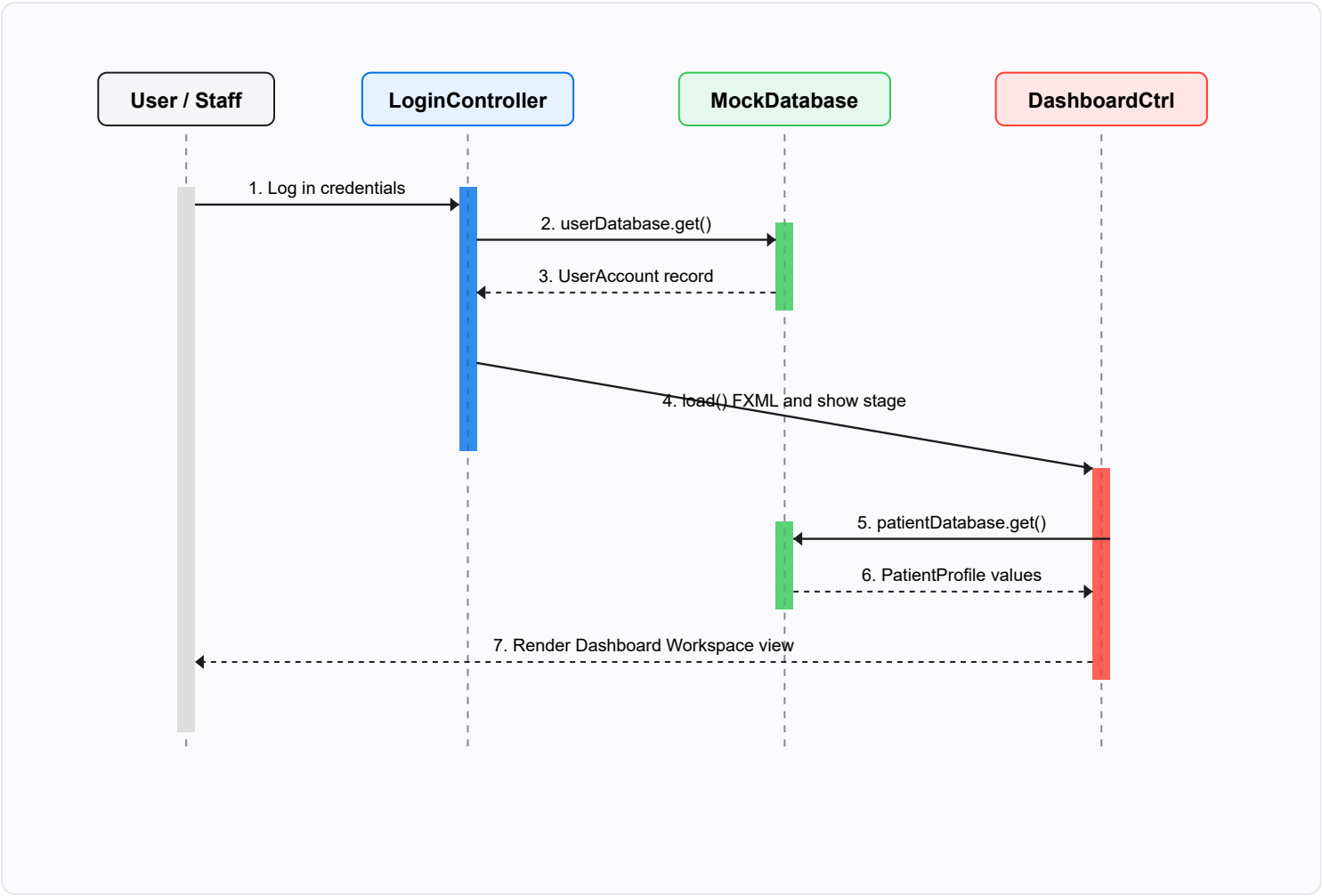
6.1 Use Case Diagram



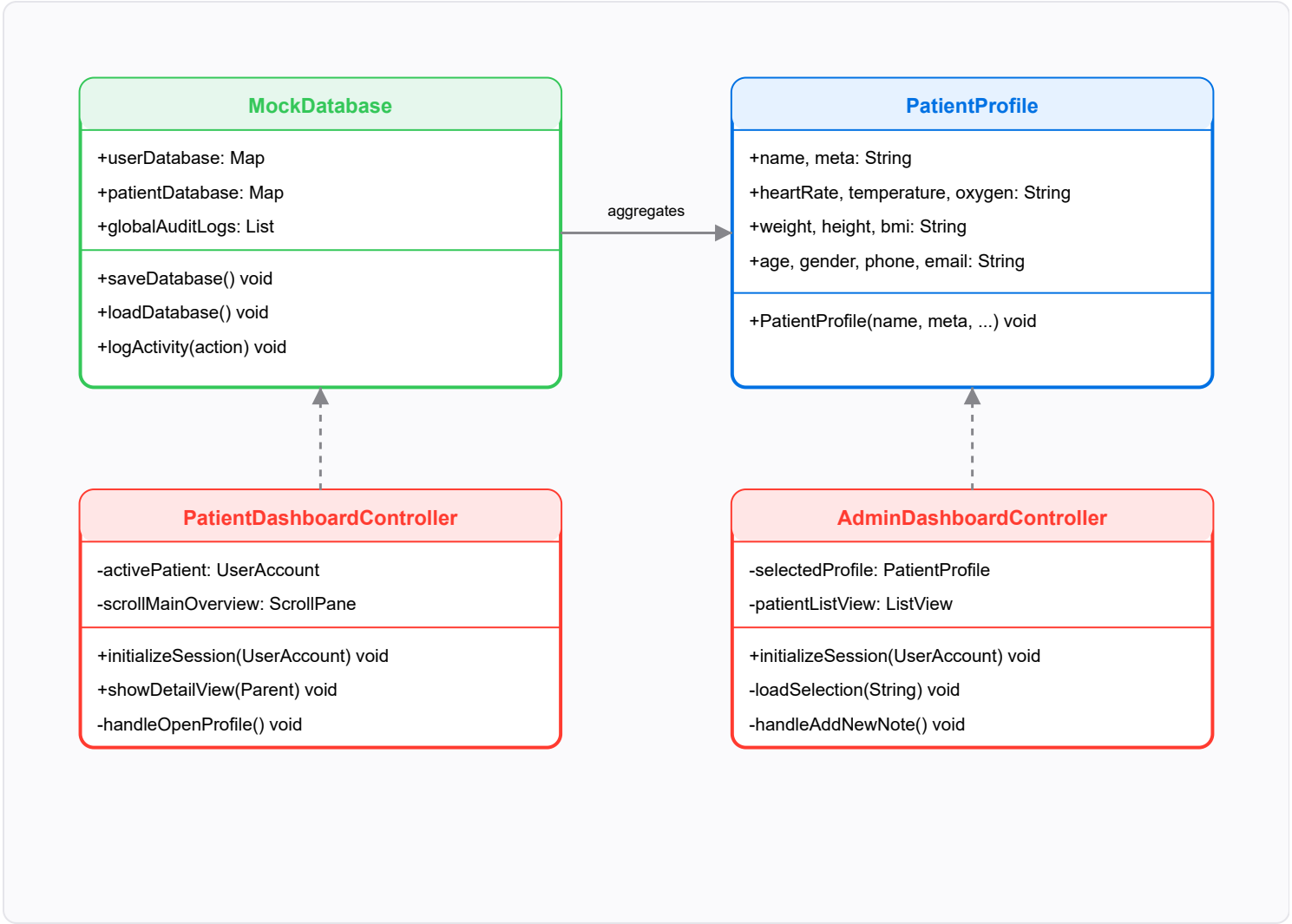
6.2 Activity Diagram (Vitals Logging Workflow)



6.3 Sequence Diagram (Authentication and Routing Flow)



6.4 Class Diagram



7. Database Design

The system implements flat-file records written to disk via ``users.db`` and ``patients.db`` in UTF-8 format. Fields are serialized using the ``|||`` delimiter, and arrays/collections are serialized with ``:::`` or ``;`` sub-delimiters.

7.1 ``users.db`` Schema

Index	Field Name	Data Type	Description
0	<code>`username`</code>	String	Primary ID key used during credential validation checks.
1	<code>`password`</code>	String	Account authentication credentials.
2	<code>`roleType`</code>	String	Session scope authorization code: <code>`PATIENT`</code> or <code>`ADMIN`</code> .
3	<code>`fullName`</code>	String	The name string display in navigation headers.
4	<code>`roleTitle`</code>	String	Clinical or support role designation (e.g. Chief Medical Officer).

7.2 ``patients.db`` Schema (Crucial Fields)

Index	Field Name	Data Type	Description / Serialized Format
0	`name`	String	The patient's full name. Map key binding.
1	`meta`	String	Location details. Legacy `"Age: XX"` prefixes are automatically stripped.
2	`heartRate`	String	Last logged resting heart rate in beats-per-minute (BPM).
3	`temperature`	String	Body temperature record in degrees Celsius (°C).
4	`oxygen`	String	Blood oxygen level in percent SpO2.
12	`bmi`	String	Dynamically calculated BMI metric value.
21	`weightHistory`	String	Comma-separated weights (e.g. `"68,67.5,67,66.8")`).
36	`clinicalNotes`	String	Notes list joined with sub-delimiter `:::`.

8. User Interface Design

The software design architecture follows **Apple's Human Interface Guidelines (HIG)** to provide a clean, modern aesthetic:

- Premium Gradients:** Colored stacked icons feature tailored linear colors, creating clear visual focus tags (e.g. Activity and Heart Rate are colored with a high-contrast rose gradient, Oxygen uses sky blue, Sleep uses an indigo-to-slate gradient).
- UI Hierarchy & Card Layouts:** Core telemetry parameters are embedded inside isolated grids, using thin borders (`-fx-border-color: #E5E5EA;`) and subtle shadow boundaries (`dropshadow`) to float components gracefully above a light gray background (`#F5F5F7`).
- Canvas & Graphic Gauges:** The Calorie Deficit Gauge is drawn dynamically on a `Canvas` node, rendering red/yellow/green zones for nutritional surplus or deficit with a rotating indicator needle based on mathematical constraints.

9. Testing and Evaluation

The system was evaluated against 10 critical validation cases to ensure functionality, input limitations, and database synchronization.

ID	Target Module	Input Test Values	Expected Result	Status
TC-01	Registration Gate	New account name, phone: `09171234567`	Accepts validation. Account seeded, metadata formatted.	PASS
TC-02	Age Validation	Age: `135` or `-4`	Triggers warning alert: "Age must be between 0 and 120."	PASS
TC-03	Phone Check	Phone: `9174567890` (10 digits)	Triggers warning: "Phone must be 11 digits and start with 09."	PASS
TC-04	Email Verification	Email: `patient432@mail.com` (with number)	Triggers warning: "Email address must not contain numbers."	PASS
TC-05	Calorie Input Range	Calories: `3200` kcal	Triggers alert: "Calories must be between 0 and 3000 kcal."	PASS
TC-06	Weight Input Range	Weight: `12.0` kg	Triggers alert: "Weight must be between 20 and 300 kg."	PASS
TC-07	BMI Metric Mapping	Height: `180` cm, Weight: `82` kg	Calculates BMI = `25.3` (Overweight badge status, Orange indicator).	PASS
TC-08	Legacy Age Removal	Metadata: `"Age: 34   Ward B, Bed 4"`	Filters age on load. Displays only `"Ward B, Bed 4"` in name subtitle.	PASS
TC-09	Telemetry Reset	Clear all user data logs	Wipes arrays. Clinical notes seeded with `"All activity and data input reset."`	PASS
TC-10	Admin Override	Clinician appends diagnostic notes	Syncs note to `patients.db` via `MockDatabase.saveDatabase()`.	PASS